REST API Documentation

Comprehensive Endpoint Catalog, Request/Response Payloads & HTTP Status Protocols

Document Title:	REST API Specification & Endpoint Integration Guide
Base URL:	http://127.0.0.1:8000 (Production / Localhost)
OpenAPI Standards:	Swagger 2.0 / OpenAPI 3.1 (/docs and /redoc)
Security Scheme:	OAuth2 Password Bearer (bearerAuth)
Data Interchange:	JSON (application/json) with Pydantic Validation

1. API Overview & Authentication Protocol

Clients authenticate by submitting credentials to POST /auth/token in form-encoded format. The issued access token must be passed in the Authorization header of subsequent requests:

```
curl -X POST "http://127.0.0.1:8000/auth/token" \
     -H "Content-Type: application/x-www-form-urlencoded" \
     -d "username=admin&password;=admin123"
# Header for protected calls:
Authorization: Bearer
```

2. Complete Endpoint Catalog

HTTP Method	Path / Endpoint	Auth Required	Description & Payload
POST	/auth/token	Public	Authenticates user; returns JWT/Bearer access token
POST	/predict	Bearer Auth	Scores single transaction; accepts TransactionInput; returns PredictionOutput
POST	/batch-predict	Bearer Auth	Scores list of transactions; automatically syncs suspicious cases to PostgreSQL
GET	/sample-197-batch	Public	Returns pre-formatted JSON payload of 197 hold-out fraud cases for testing
GET	/alerts	Bearer Auth	Retrieves stored fraud alerts with optional query parameter ?status=
GET	/alerts/{id}	Bearer Auth	Fetches detailed record of a single alert by primary ID
PUT	/alerts/{id}/status	Bearer Auth	Updates alert status (New ➔ Under Review ➔ Resolved)
DELETE	/alerts/{id}	Bearer Auth (Admin)	Permanently deletes an alert record from database
GET	/alerts-summary	Bearer Auth	Returns aggregate counts of total, new, under review, and resolved alerts
GET	/health	Public	Health check probe confirming API, ML model, and PostgreSQL connectivity
GET	/model/monitoring-stats	Bearer Auth	Returns live operational model metrics, active threshold (0.42), and version

Financial Fraud Detection System

3. Request & Response Payload Examples

Single Transaction Scoring (`POST /predict`):

```
// Request Body (JSON)
{
  "Transaction_ID": "T5438",
  "Timestamp": "2026-10-14 03:15:22",
  "Sender_Account": "ACC-9921",
  "Receiver_Account": "ACC-1044",
  "Amount": 3379.23,
  "Type": "Transfer",
  "Location": "New York",
  "Account_Balance": 12500.00,
  "Device_ID": "DEV-3392",
  "IP_Address": "192.168.1.105",
  "User_Device_Recognition": 0,
  "Known_Threat_Flag": 1,
  "Login_Location_Match": 0,
  "Spending_Pattern_Deviation": 2.45
}
// Response Body (200 OK)
{
  "transaction_id": "T5438",
  "fraud_probability": 0.4631,
  "prediction": "Suspicious",
  "risk_score": 46.31,
  "risk_level": "High",
  "alert_id": 104,
  "message": "Transaction flagged as suspicious and recorded in alert management."
}
```

4. Standard HTTP Response Codes

Status Code	HTTP Reason	Condition
200 OK	Success	Request processed successfully; prediction or data returned
400 Bad Request	Client Error	Invalid query parameter or malformed status transition
401 Unauthorized	Authentication Required	Missing, invalid, or expired Bearer token in header
404 Not Found	Resource Not Found	Specified alert ID does not exist in database
422 Unprocessable Entity	Validation Error	Pydantic validation failed; field missing or incorrect data type
500 Internal Error	Server Error	Database connection lost or unhandled exception